

Computer Vision Practical Report:

BLACK HAT TECHNIQUE

1. AIM

To compute the Black-Hat transform (the difference between the morphological closing of an image and the original image) to isolate structures and features that are darker than their surrounding background using `cv2.morphologyEx()` in OpenCV, display the results, and save the output image.

2. PROCEDURE

- Load the input image using `cv2.imread()` and convert it to grayscale using `cv2.cvtColor()`.
- Define a spatial structuring element (kernel) using `cv2.getStructuringElement()` (e.g., rectangular or elliptical).
- Apply the Black-Hat transform using `cv2.morphologyEx()` with `cv2.MORPH_BLACKHAT`.
- Display the original and Black-Hat transformed images side-by-side using Matplotlib.
- Save the output image to disk using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load image
img_bgr = cv2.imread('E35-IMG.jpg')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

# 2. Define Structuring Element (9x9 kernel to isolate small dark spots)
kernel = np.ones((9, 9), np.uint8)

# 3. Perform Black Hat Operation (Closing - Original)
blackhat = cv2.morphologyEx(img_rgb, cv2.MORPH_BLACKHAT, kernel)
```

4. Display side-by-side on a SINGLE PAGE

```
fig, axes = plt.subplots(1, 2, figsize=(8, 4))
```

```
axes[0].imshow(img_rgb)
```

```
axes[0].set_title('Original Image')
```

```
axes[0].axis('off')
```

```
axes[1].imshow(blackhat)
```

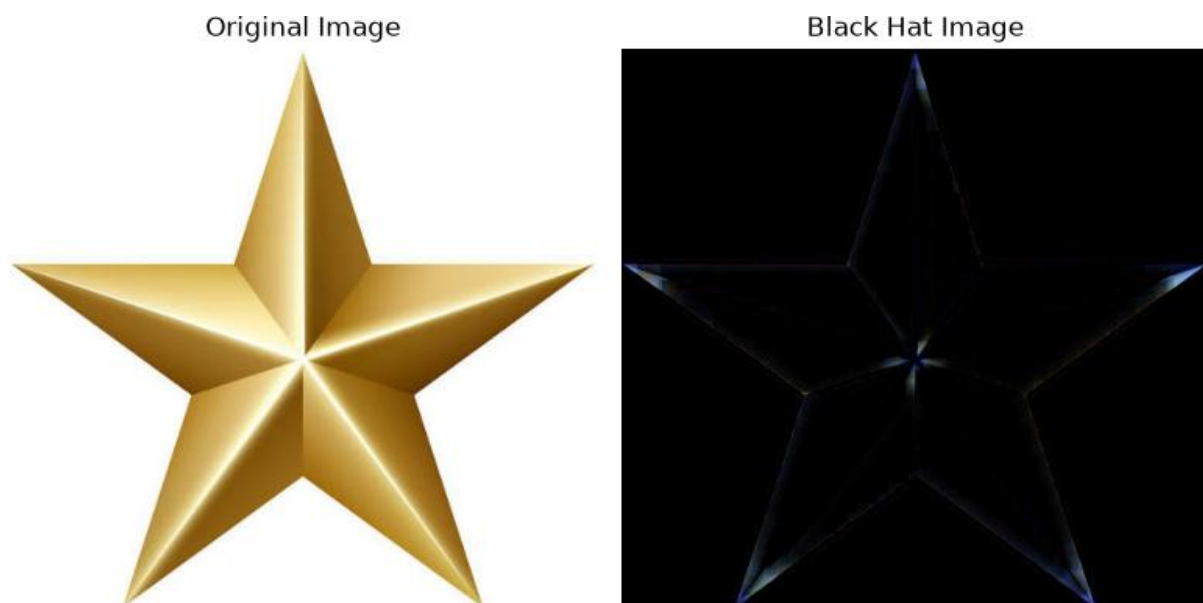
```
axes[1].set_title('Black Hat Image')
```

```
axes[1].axis('off')
```

```
plt.tight_layout()
```

```
plt.show()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.

